

Job Hunt

Project Description : A smart recruitment platform designed to revolutionize the hiring process. It seamlessly connects talented job seekers with leading companies through an intuitive and user-friendly interface, powered by intelligent matching algorithms.





Our Team

Driving innovation and excellence in recruitment technology.

Project Team Members

- Amr Mohamed (2202388)
- Sondos Ahmed (2202343)
- Nada Abd Elsattar (2202310)
- Mariam Samir (2202386)
- Mohamed Emad (2202417)
- Ahmed Sobhy (2202234)
- Ziad Magdy (2101918)
- Mariam Tarek (2202384)

Supervisors

- Dr. Mohamed Saber
- Eng. Soaad Mustafa

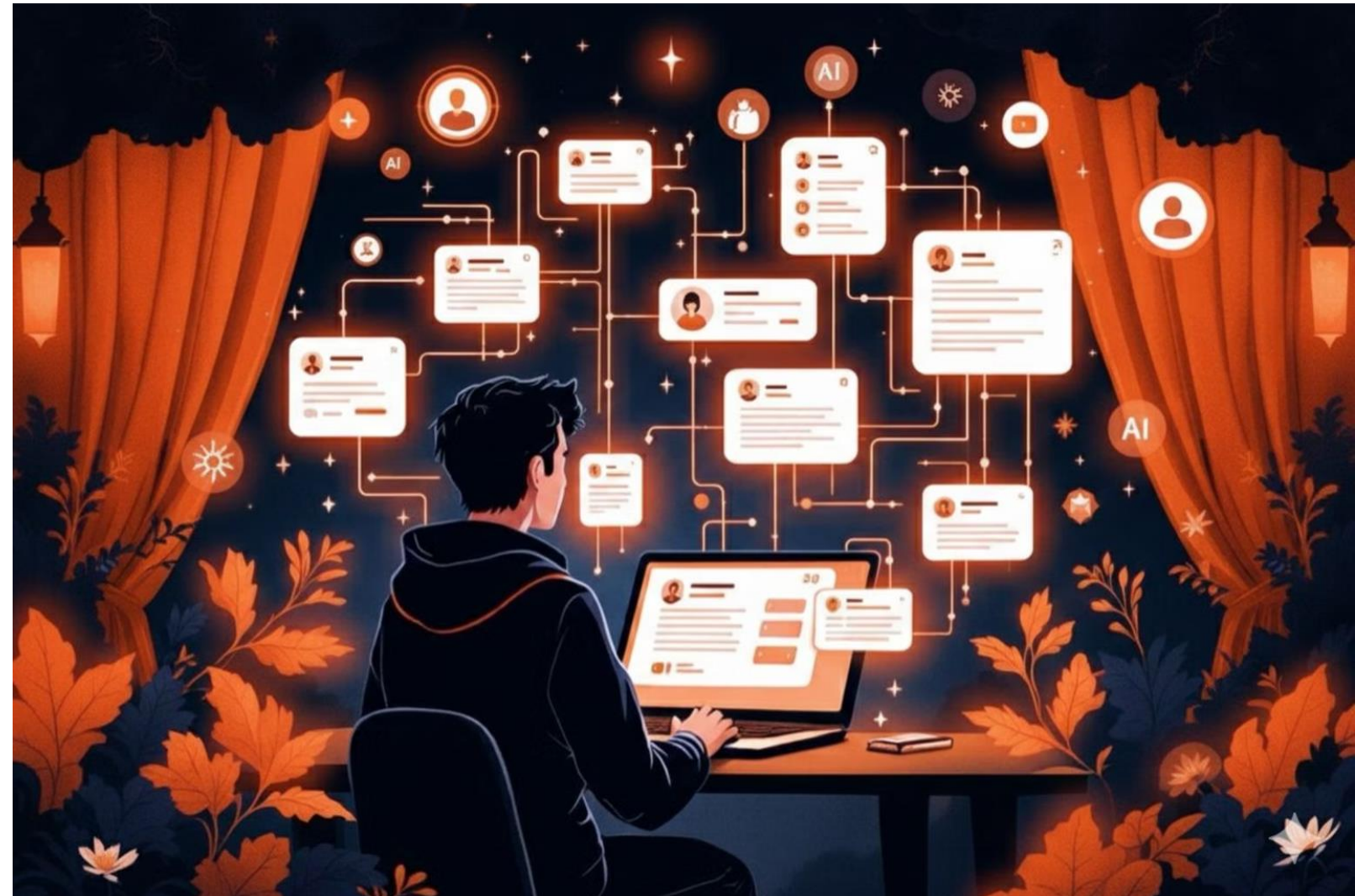
Presentation Outline

Here's what we'll cover in today's discussion about the Job Hunt platform:

01	02	03
Introduction	Problem Definition	Related Work
Setting the stage for our innovative recruitment platform.	Understanding the challenges Job Hunt aims to solve.	Analysis of existing solutions and market landscape.
04	05	
Proposed Solution	System Design	
Detailed look at the features and benefits of Job Hunt.	Overview of the architecture and technical implementation.	

Introduction

- Job Hunt is a web-based platform that helps users find suitable job opportunities in a simple, efficient, and organized way.
- It connects job seekers with relevant job listings by matching their information with job requirements, increasing the chance of finding the right role faster.
- The platform provides smart and personalized recommendations to save time and reduce the effort of traditional job searching.





What is Job Hunt?

Project Idea

Job Hunt is an intelligent job searching system that revolutionizes how users connect with opportunities. Instead of manually sifting through thousands of job posts, our platform guides users directly to roles that fit their profiles through advanced matching algorithms.



Effortless CV Upload

Seamlessly upload and manage your curriculum vitae, making your profile instantly accessible to potential employers.



Explore Diverse Opportunities

Discover and browse a vast array of job listings, filtered and tailored to your specific skills and career preferences.



Personalized Recommendations

Receive intelligent, AI-driven job suggestions that perfectly align with your experience and aspirations, saving you time and effort.

Why Job Hunt?

Job Hunt simplifies the entire recruitment ecosystem, offering significant advantages for both job seekers and employers.

1

For Job Seekers

Easily upload your CV and receive personalized job recommendations, cutting through the noise of irrelevant listings.

- Access tailored opportunities
- Streamlined application process
- Save valuable time

2

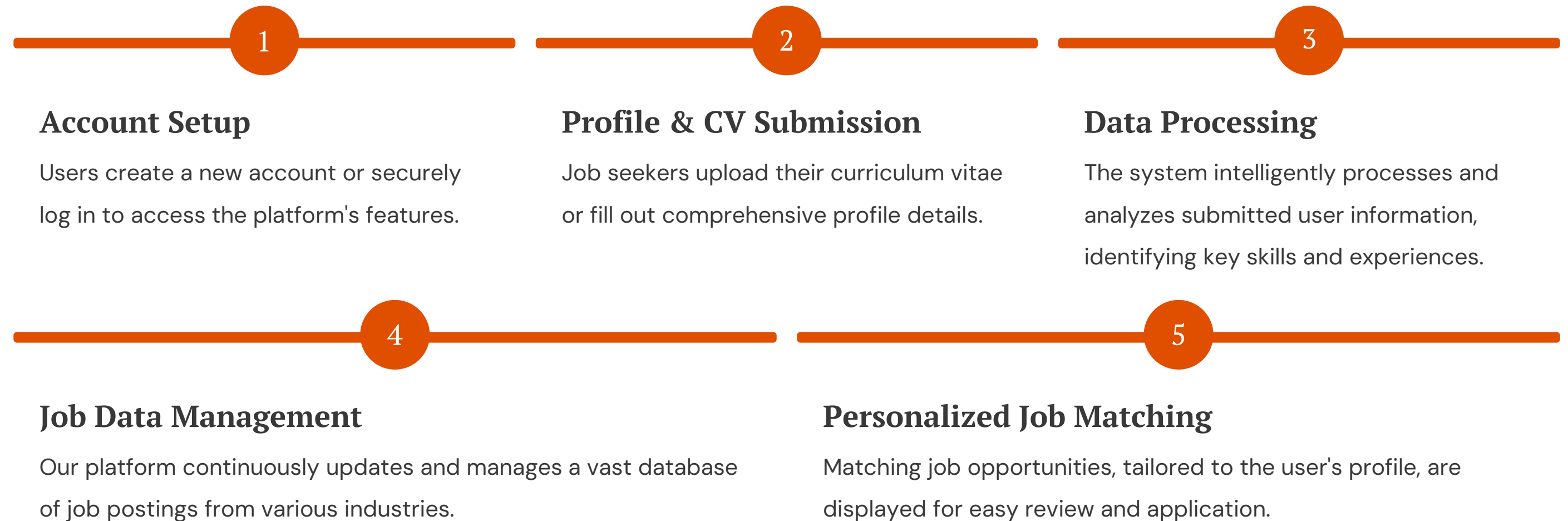
For Employers

Efficiently identify top talent that perfectly matches your criteria, powered by intelligent matching algorithms.

- Faster candidate sourcing
- Higher quality matches
- Reduced recruitment costs

How Job Hunt Works

System Workflow



This workflow ensures accurate and meaningful job suggestions, streamlining the path to ideal career opportunities.

Technologies & Tools Used

Key Components Powering Job Hunt

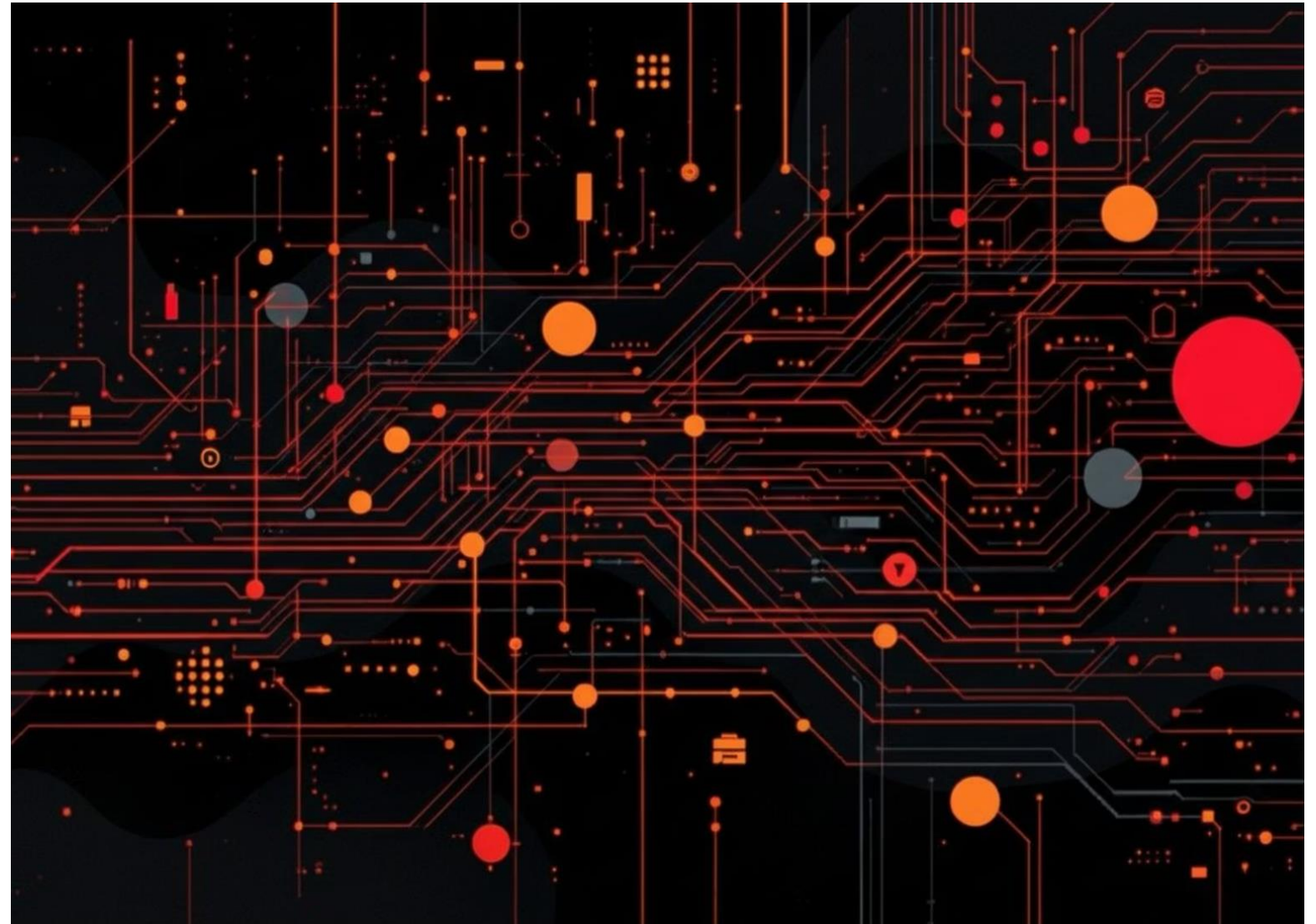
Web Application: Robust frontend and backend development for a seamless user experience.

CV Parsing: Advanced text extraction capabilities for detailed profile analysis.

Database Management: Secure storage of job listings and comprehensive user data.

Recommendation Logic: Intelligent algorithms to precisely match users with relevant jobs.

User Interface: Intuitive and user-friendly design for effortless navigation.



Problem Definition

The Problem

- The modern job search is often a daunting, complex, and time-consuming process.
- Job seekers struggle to navigate overwhelming listings and find opportunities that truly align with their unique skills and experience due to poor filtering.
- This inefficiency leads to frequent applications to irrelevant roles, resulting in frustrating rejections and prolonged job hunts.



Current Challenges in Job Searching

Job seekers today face numerous obstacles that make finding the right opportunity a daunting task:

- **Overwhelming Listings:** A sheer volume of job posts often lack personalization, making it hard to sift through relevant options.
- **Skill Mismatch:** Difficulty in precisely aligning one's skills and experience with specific job requirements, leading to missed opportunities.
- **Manual & Disorganized Process:** Traditional job searching methods are often chaotic and inefficient, consuming valuable time and energy.
- **Irrelevant Applications:** Valuable time is frequently wasted applying to roles that are not a good fit, resulting in low success rates.
- **Lack of Guidance:** Fresh graduates often struggle with a clear path forward in their career search, lacking tailored advice and direction.

These prevalent issues highlight the urgent need for a more streamlined, intelligent, and personalized job hunting experience.



Impact of the Problem

The inefficiencies in the traditional job search process create significant challenges for both job seekers and employers, widening the gap between talent and opportunity.

For Job Seekers

Missed Opportunities: Suitable roles are overlooked due to overwhelming listings and poor filtering.

Low Success Rate: Increased effort yields minimal results, leading to frustration and prolonged job hunts.

Poor User Experience: Traditional platforms often lack personalization, making the process daunting and inefficient.

For Companies

Unqualified Applications: Recruiters are flooded with resumes that don't match job requirements, wasting valuable time.

Increased Costs: Longer hiring cycles and the need for extensive screening processes drive up recruitment expenses.

Talent Gap: Difficulty in efficiently identifying and attracting the right talent, impacting business growth and innovation.

This creates a critical disconnect that hinders career progression and organizational success, underscoring the urgent need for an effective, intelligent solution like Job Hunt.

Related Work Overview

Existing Job Search Systems

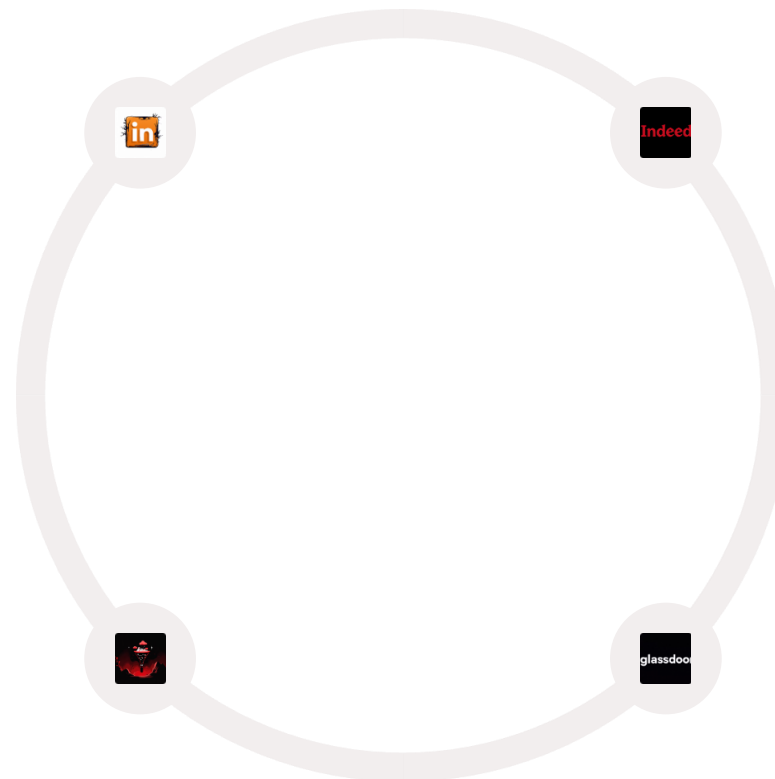
Several established platforms aim to connect job seekers with opportunities, but often present inherent limitations in personalization and efficiency.

LinkedIn

Primarily a professional networking site, offering extensive job listings. Its search primarily relies on keyword matching and user-filtered criteria.

Limited Personalization

These systems often overwhelm users with volume and depend heavily on manual filtering, leading to a suboptimal experience for tailored job discovery.



Indeed

One of the largest job boards, aggregating postings from numerous sources. It provides a wide reach but often lacks deep personalization beyond basic filters.

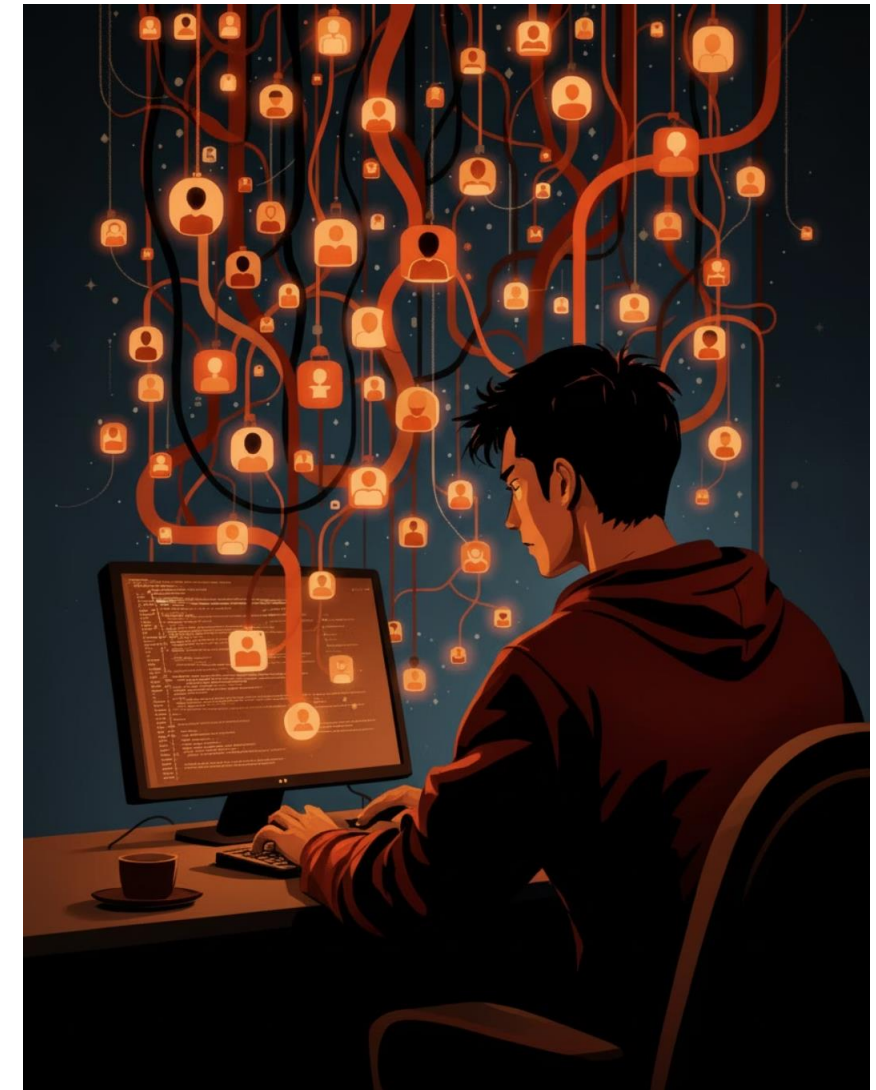
Glassdoor

Combines job search with company reviews and salary data. While valuable for research, its core job matching still largely depends on explicit search terms.

Limitations of Existing Solutions

Current job search platforms, while widely used, suffer from several critical shortcomings that hinder an efficient job hunt:

- **Keyword-Based Matching:** Often relies on superficial keyword matching instead of genuine skill and experience alignment.
- **Manual Filtering Burden:** Users are left to manually sift through countless irrelevant listings, consuming valuable time.
- **Limited CV Analysis:** CV parsing is often rudimentary or non-existent, missing deep insights into a candidate's profile.
- **Absence of Recommendation Systems:** Lacks clear, personalized job recommendation systems for job seekers.
- **Inadequate for Fresh Graduates:** Not well-suited for fresh graduates or those with limited experience, who require more structured guidance.



As a result, many users still struggle to find suitable job opportunities efficiently, leading to prolonged searches and frustration.

Research-Based Approaches

Previous Research Efforts

Previous research has proposed various job recommendation systems using diverse methodologies:

- Rule-based systems
- Traditional machine learning techniques
- Content-based filtering approaches

However, many of these systems suffer from critical limitations, hindering their effectiveness and user adoption:

Single Feature Focus: Often concentrate on a solitary attribute, such as skills or experience, neglecting a holistic candidate profile.

Lack of CV Analysis: Typically do not support direct CV upload and in-depth analysis, missing crucial candidate insights.

Incomplete Implementations: Not fully developed into comprehensive, user-friendly applications that can be readily adopted by job seekers.

Research Gap: Unmet Needs

Despite the proliferation of job platforms and extensive research, critical functionalities are still missing to create a truly effective job hunting experience:

Automated CV Analysis :

A system capable of intelligently parsing and understanding user CVs beyond simple keyword extraction.

Skills-Based Recommendations :

A sophisticated recommendation engine that precisely matches job seekers with roles based on their deep skills and experience.

Interactive User Interface :

An intuitive and engaging user experience that simplifies the complex job search process.

End-to-End Solution :

A comprehensive platform offering support throughout the entire job hunting journey, not just job discovery.

This significant gap in existing solutions directly motivated the development of the Job Hunt System.

Proposed Solution: Job Hunt Overview

To address the limitations of existing job search platforms, we propose **Job Hunt**, a smart and user-friendly job recommendation system designed to revolutionize the job seeking experience.



Intelligent Recommendation

Job Hunt utilizes advanced algorithms to provide highly personalized job recommendations, moving beyond simple keyword matching.



Advanced CV Analysis

Our system intelligently parses and understands job seekers' CVs, extracting deep insights into their skills and experience.



Streamlined Process

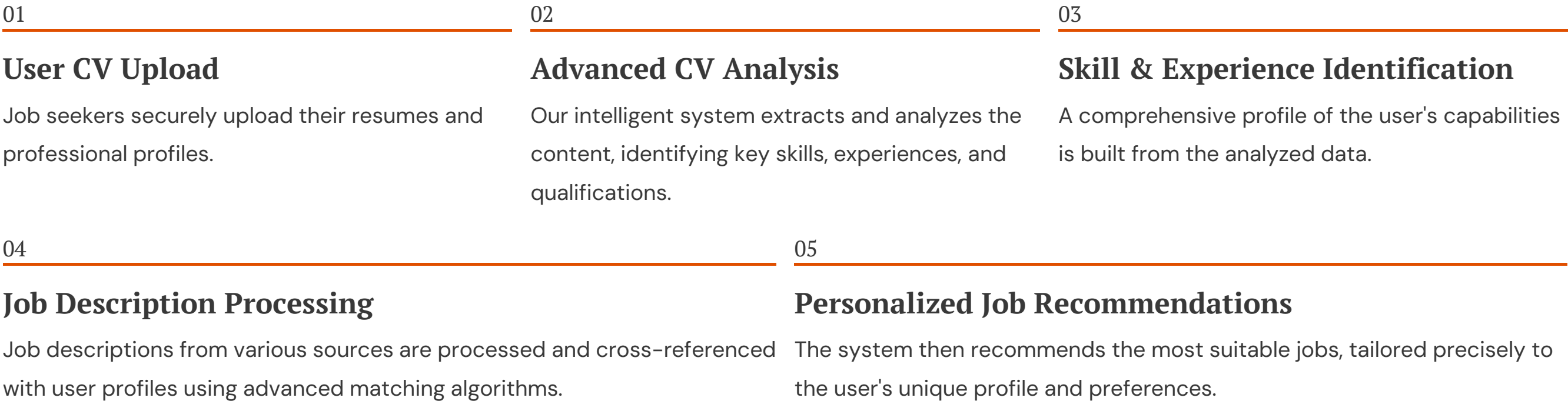
By precisely matching candidates with relevant roles, Job Hunt significantly reduces the time and effort required for successful job hunting.

Job Hunt aims to simplify the job searching process and reduce the time and effort required by users, fostering more efficient and satisfying career transitions.

How the System Works

System Workflow

The proposed system works through the following streamlined steps, ensuring a personalized and efficient job search experience:



This methodical process provides highly personalized job recommendations, significantly enhancing the relevance and success rate for each user.

Key Features of the Proposed System

Main Features



Automated CV Analysis

Intelligently parses resumes to extract deep insights into skills and experience.



Skill-Based Job Matching

Precise algorithms align job seekers with roles based on their specific capabilities.



Personalized Recommendations

Tailored job suggestions driven by individual profiles and evolving preferences.



Simple User Interface

An intuitive and engaging experience that streamlines the entire job search process.

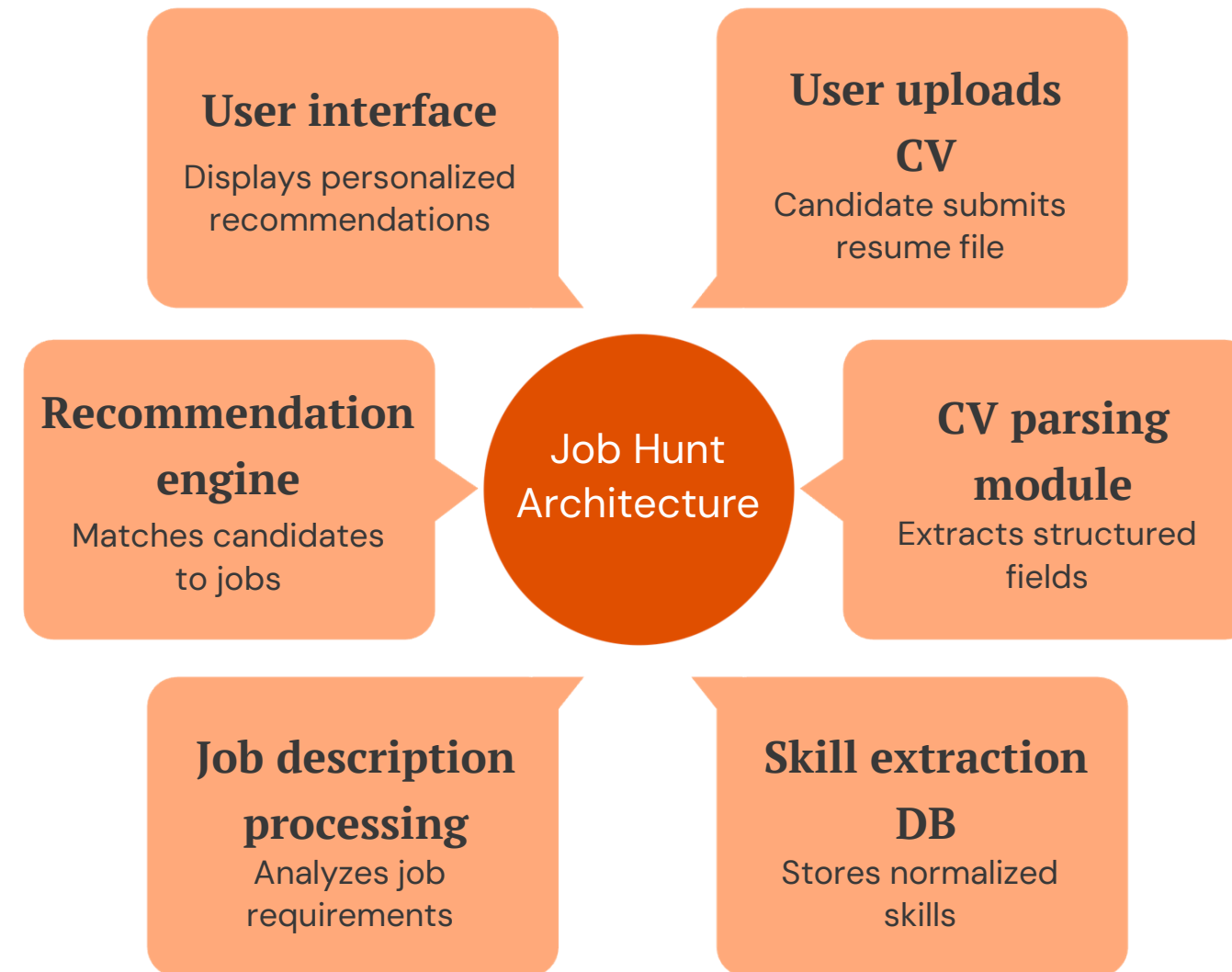


Inclusive for All Levels

Designed to benefit both fresh graduates and seasoned professionals alike.

System Design

A high-level overview of the Job Hunt system's architecture, detailing its core components and their interactions.



This architecture ensures seamless data flow from raw CVs to personalized job recommendations, leveraging intelligent processing at each stage.

Project Roadmap

A strategic overview of the Job Hunt system's development phases, outlining key milestones from inception to deployment.

01

Requirement Analysis

Defining functional and non-functional system requirements.

03

Backend & AI Development

Implementing server-side logic and AI/ML algorithms.

05

Integration & Testing

Combining modules and rigorous testing for functionality.

This structured approach ensures efficient development and a robust final product.

02

System Design

Designing architecture, database, and core components.

04

Frontend Development

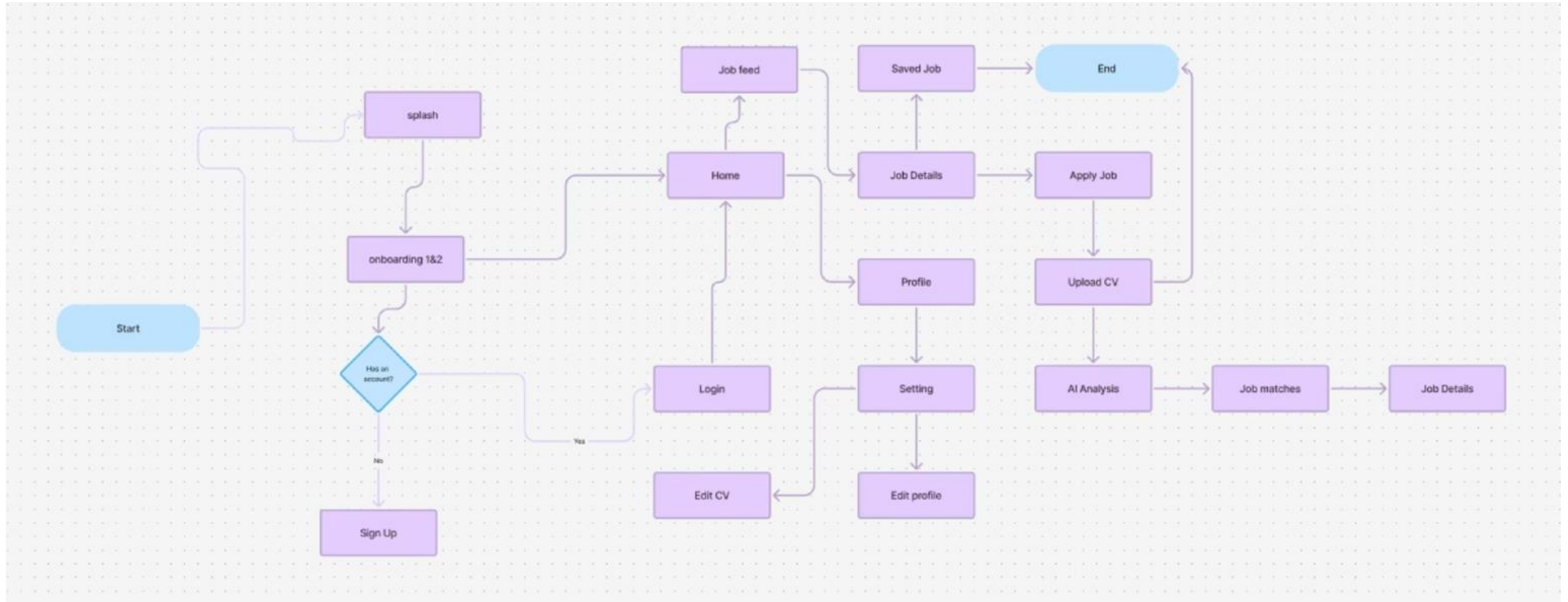
Building the intuitive user interface and experience.

06

Deployment & Future Work

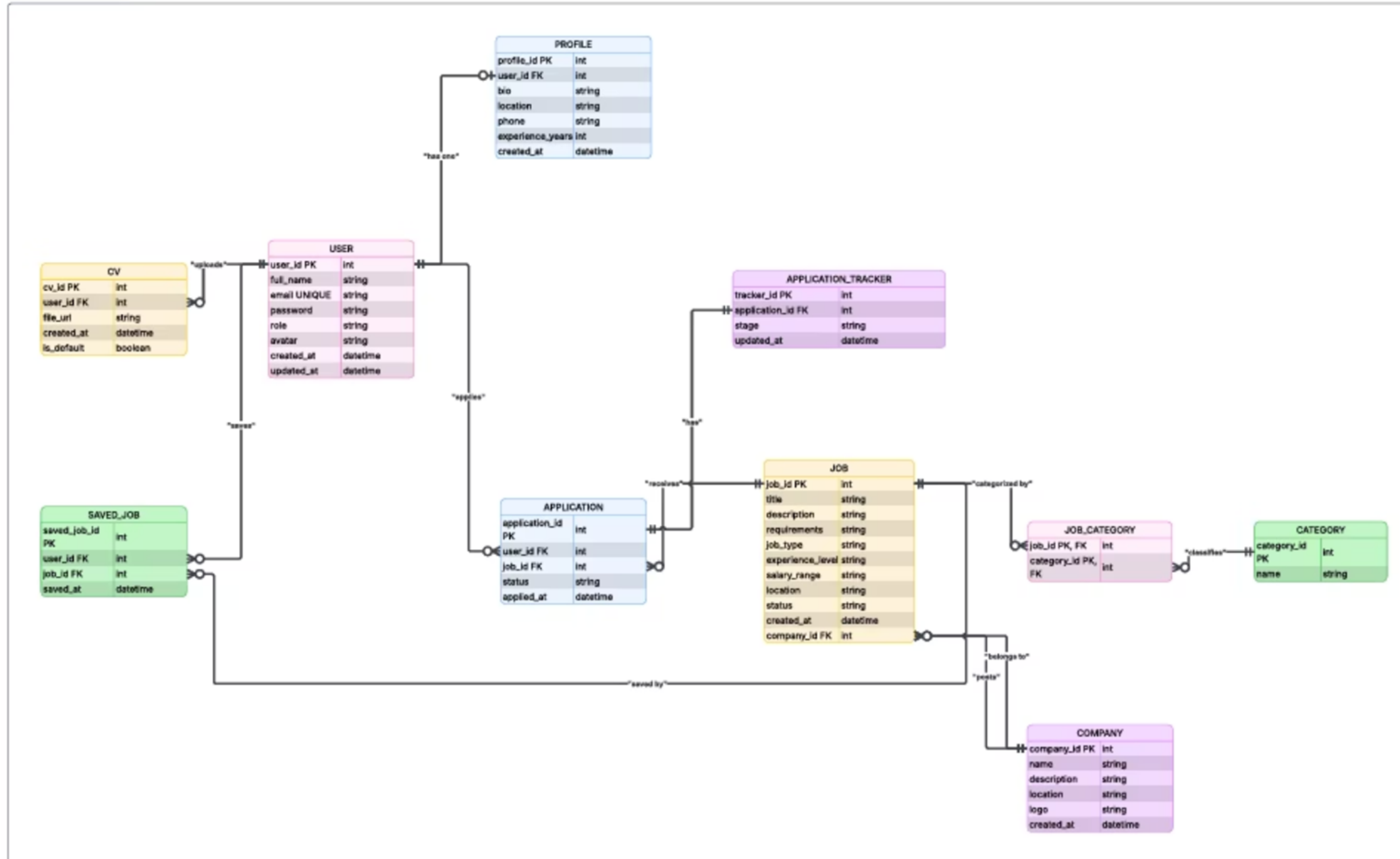
Launching the system and planning continuous improvements.

User Flow

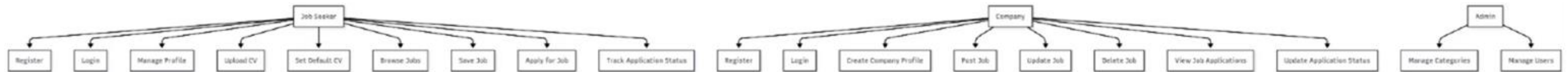


ERD Diagram + System Database Schema

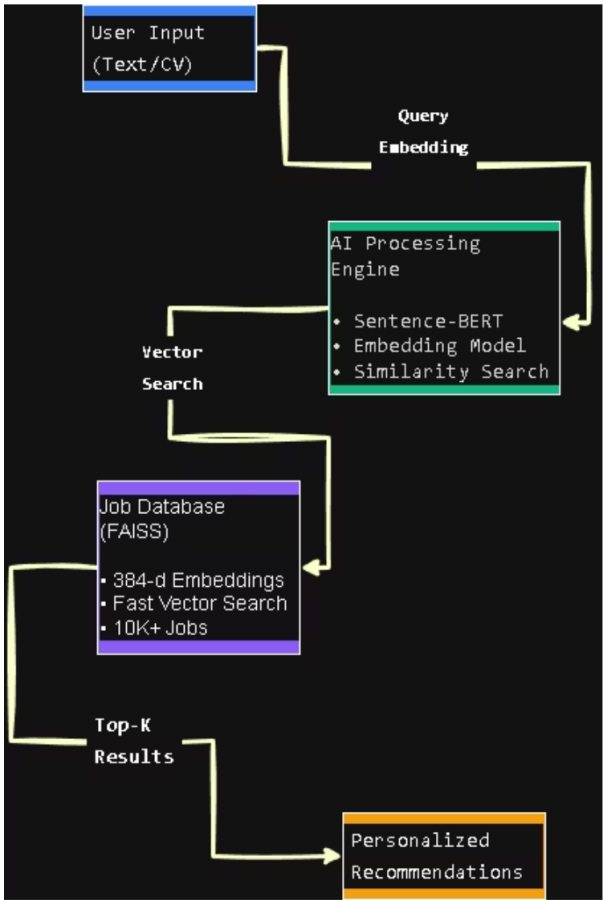
Job Hunt ERD



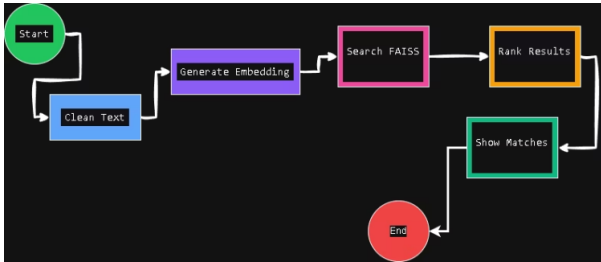
Use Case Diagram



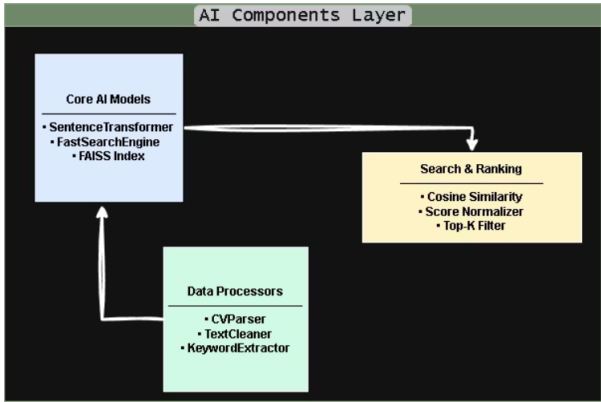
Ai diagrams



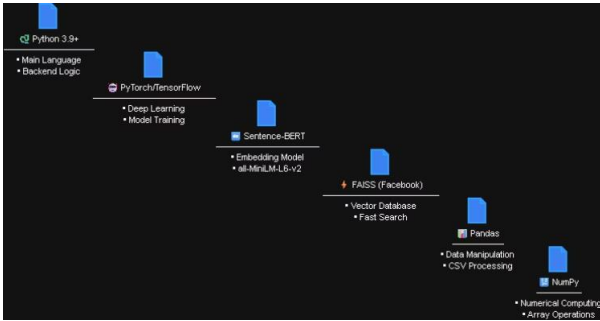
Ai System Overview



Data Flow Diagram



Component
Architecture



Technical Stack

Implementation

Backend Technologies :

Framework :

Node.js – Express.js

API Style :

REST

Database :

SQL Server / MongoDB

Authentication :

JWT – Role-Based Authorization – Password Hashing (bcrypt)

Validation :

Joi / express-validator

File Upload :

Multer – LocalStorage / Cloud

Caching :

In-Memory / Redis

Security :

Helmet – CORS – Rate Limiting

Logging :

Morgan / Winston

Monitoring :

Health Checks

Deployment :

Environment Variables – PM2 – (Hosting Provider)

Frontend Technologies :

Framework :

React.js

State Management :

Context API

Styling :

Tailwind CSS

HTTP Client :

Fetch API

Routing :

React Router DOM

Form Handling :

React Hook Form – Client-side Validation

File Upload :

FormData API – FileReader API (Preview)

DOM Manipulation :

Direct (useRef, Vanilla JS)

Build Tool :

Vite / CRA

Architecture :

Component-Based – SPA

AI Service Technologies :

Framework :

Flask (Python)

ML Library :

scikit-learn

Text Vectorization :

TF-IDF

Similarity Measure :

Cosine Similarity

Skill Extraction :

Custom Keyword Matching

Scoring Formula :

$(0.6 \times \text{Skill}) + (0.4 \times \text{Text})$

Data Handling :

Pandas – NumPy – CSV

API Endpoints :

/recommend (POST JSON) – /recommend-file (POST file) – /health (GET)

Deployment :

Localhost / PythonAnywhere / Docker (optional)

Other Tools & Practices :

Version Control :

Git – GitHub

API Testing :

Postman

Environment Config :

dotenv

Code Quality :

ESLint – Prettier (optional)



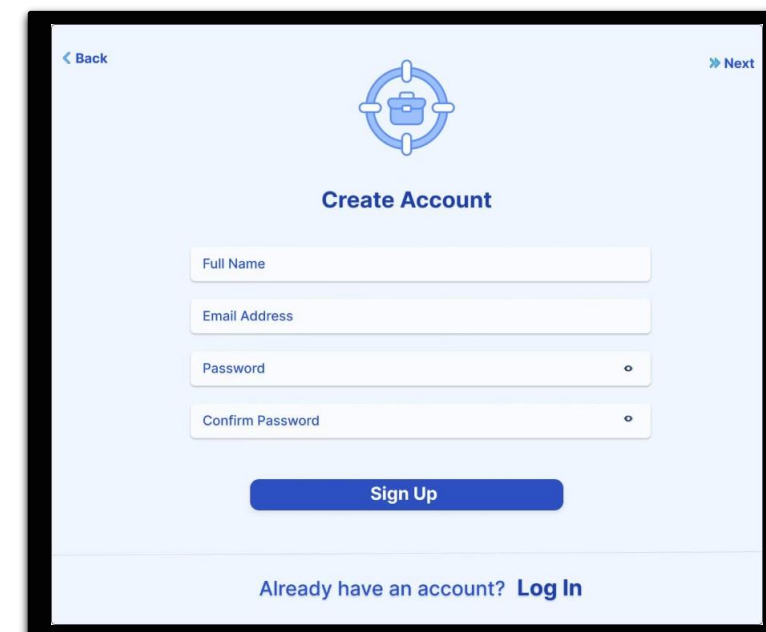
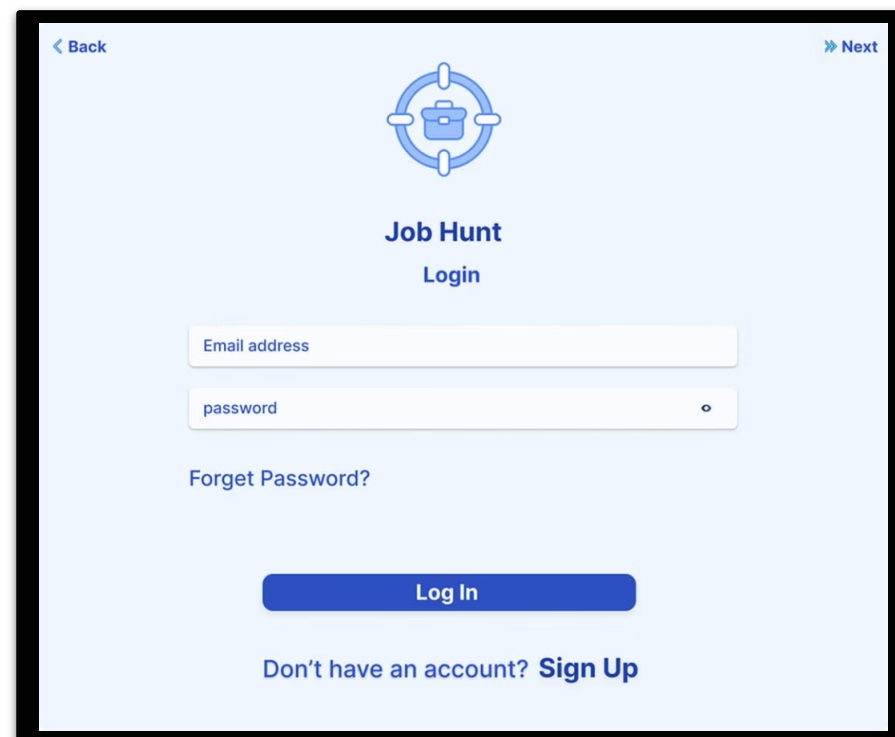
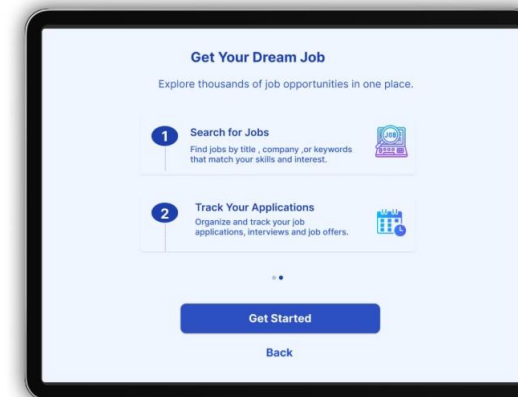
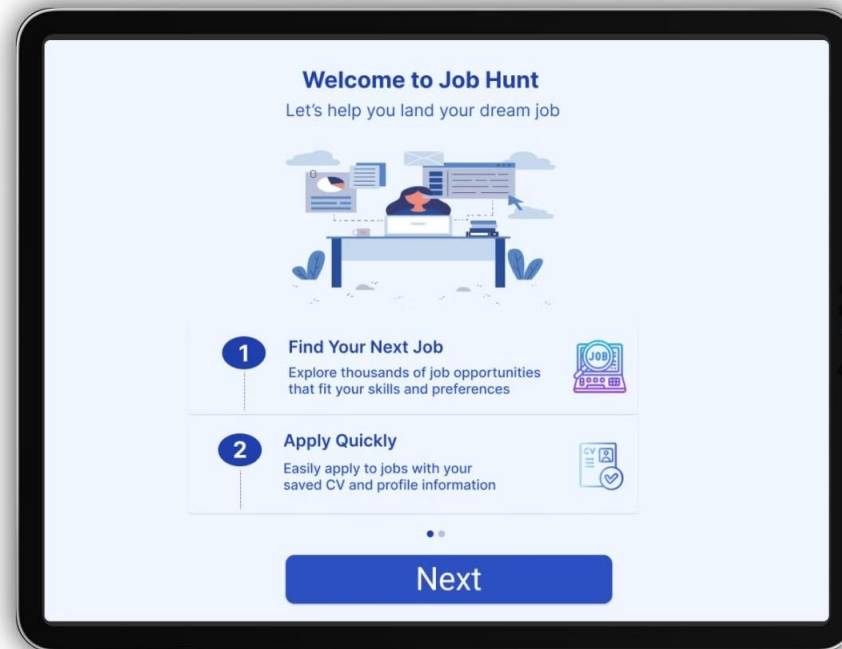


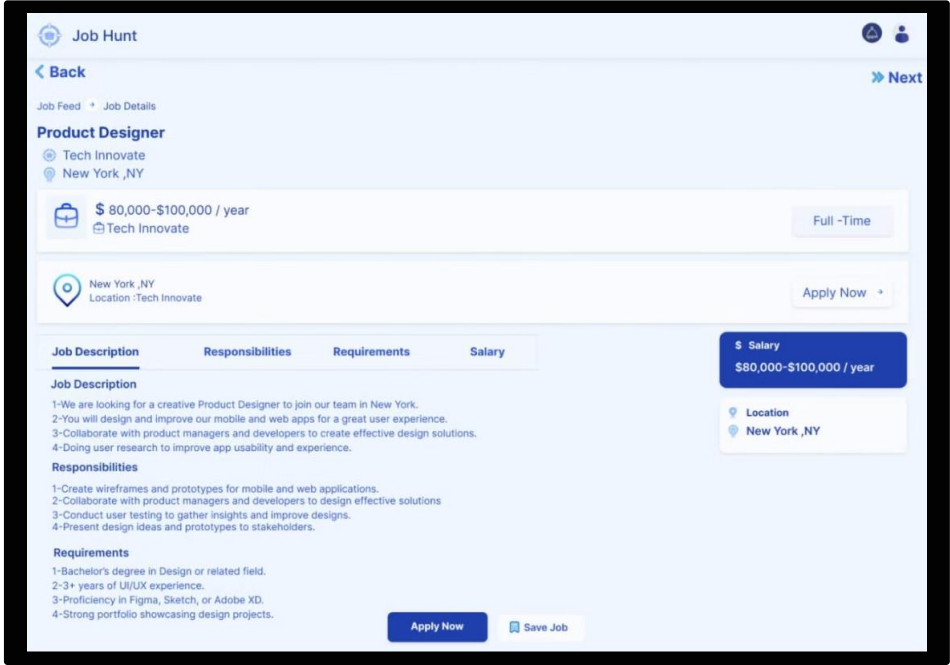
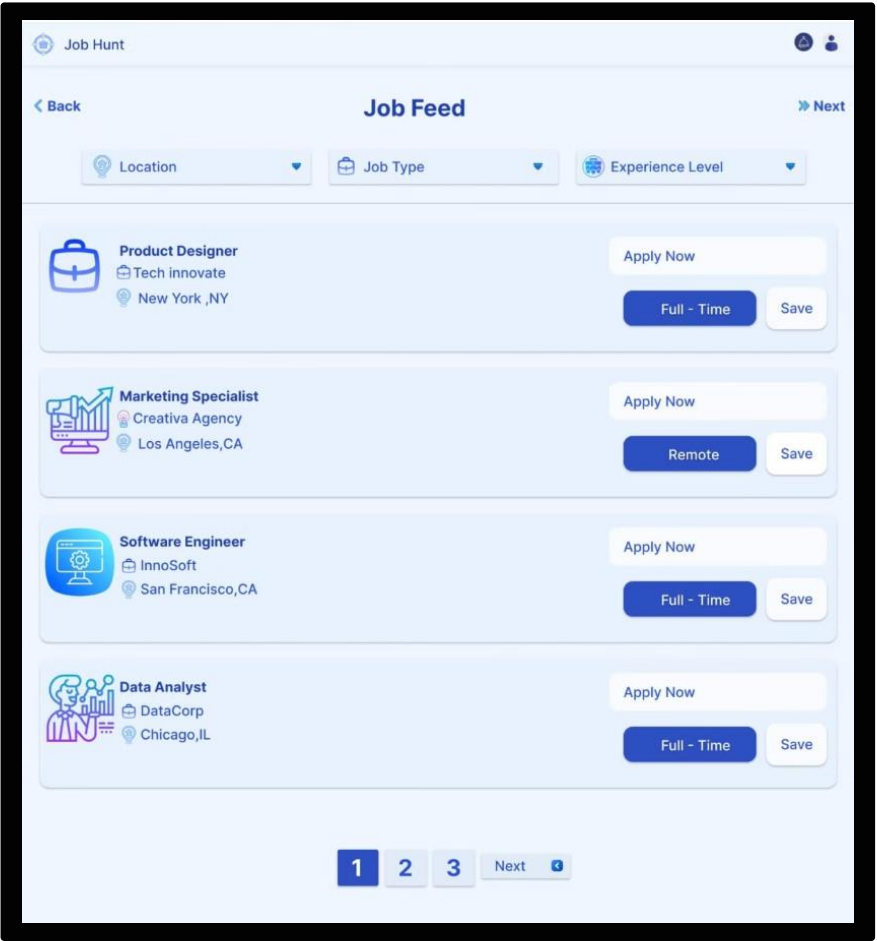
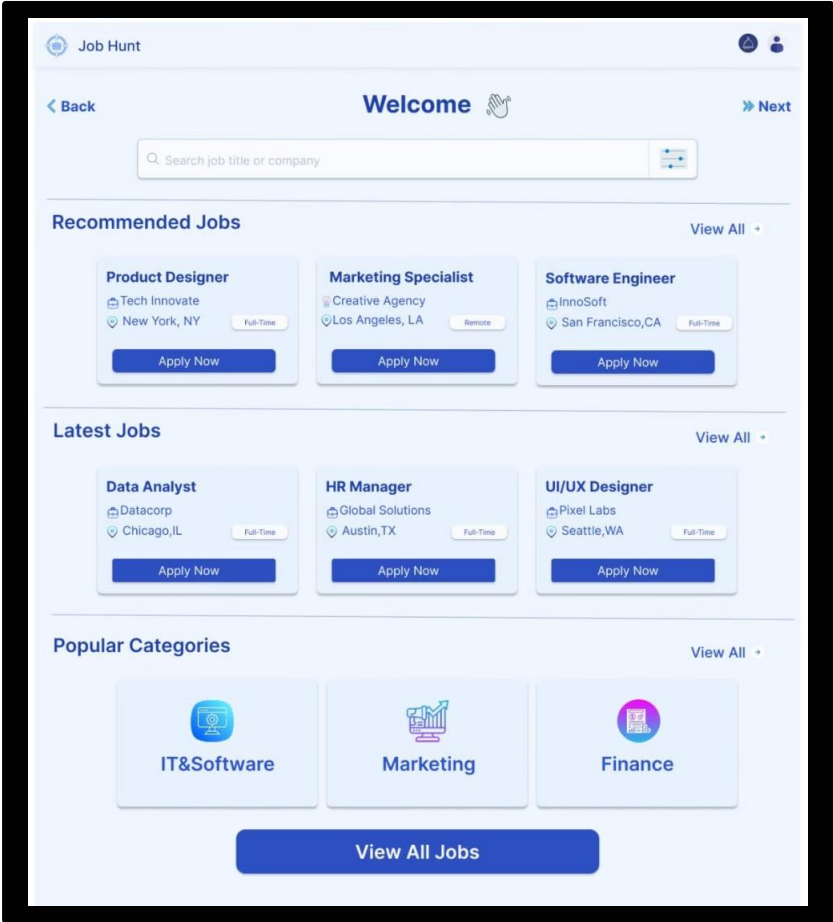
Job Hunt

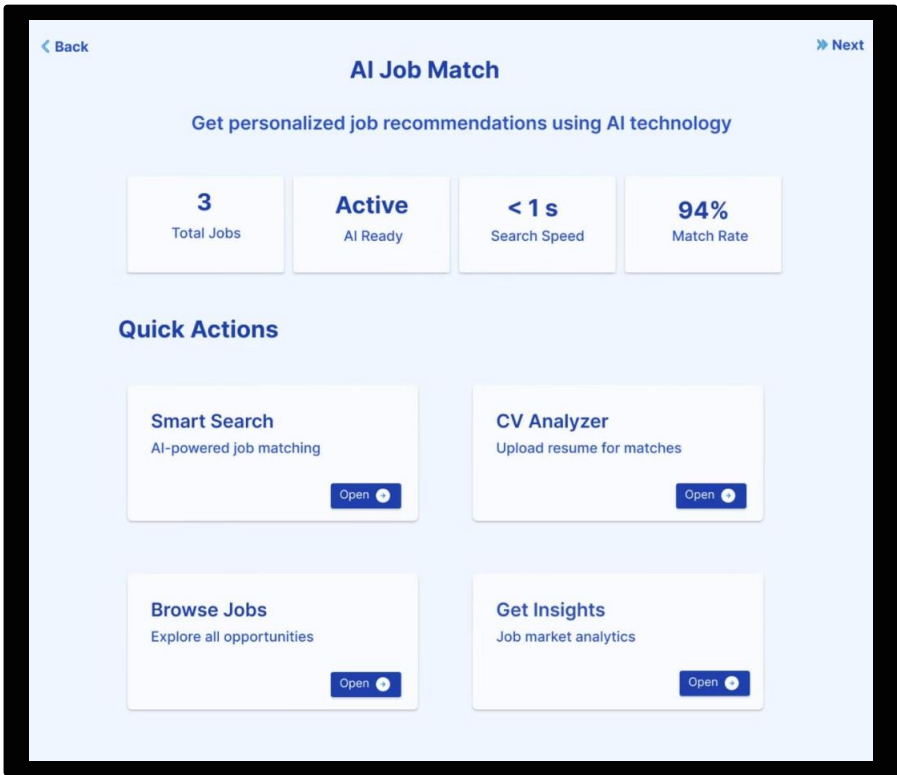
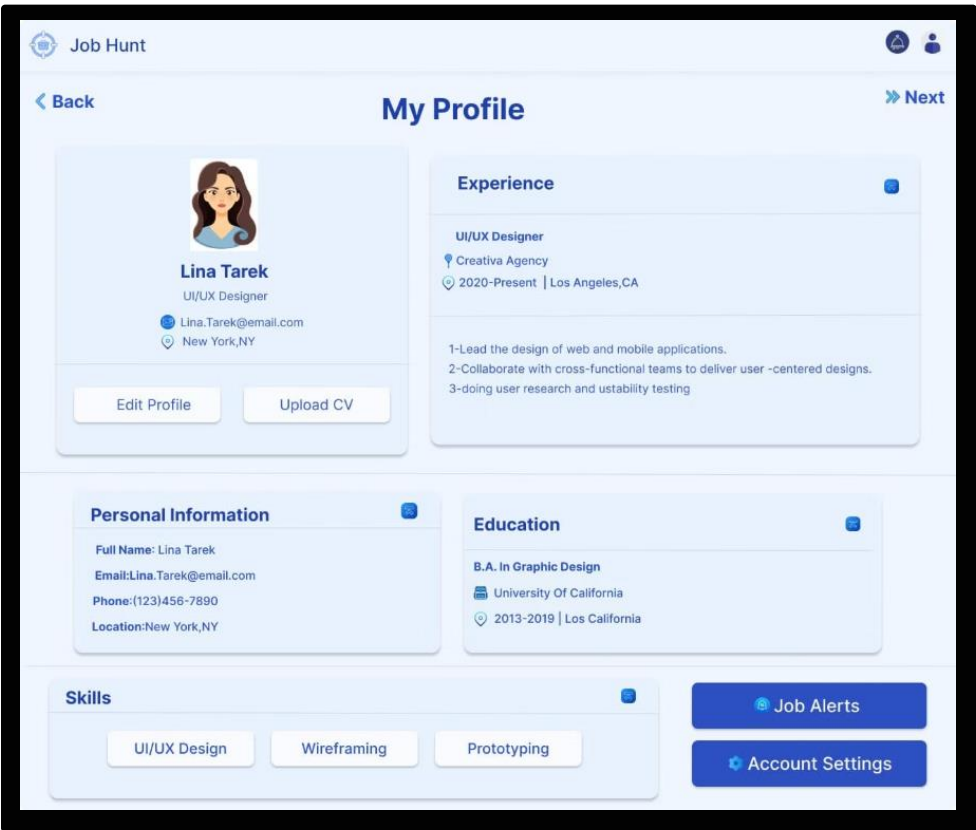
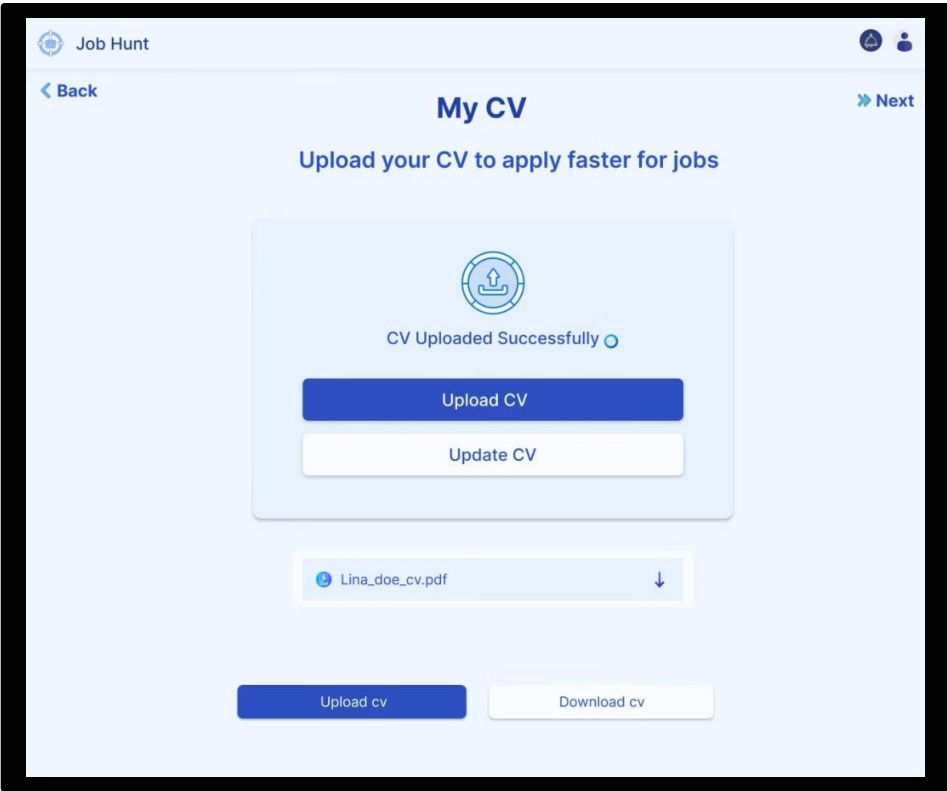
Find your dream job easily

Job Hunt

Start :







Thank you

